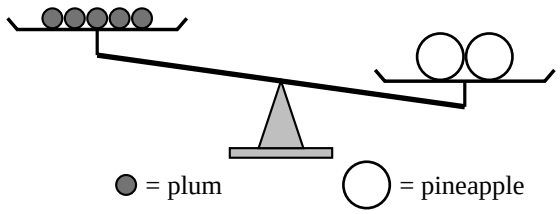
Math Kangaroo Practice

Grade 3-4 · 20 questions · harder · seed 77718

Part A: Each question is worth 3 points

1. On one plate of a balance there are 5 plums and on the other there are 2 pineapples. When we put one more pineapple, exactly like the others, on the plate with the plums, the balance is level.



One pineapple weighs the same as how many plums?

A. 5 plums ***B.*** 6 plums ***C.*** 7 plums ***D.*** 8 plums ***E.*** 10 plums

2. Lucas, Hana, Leo, Fabienne and Sofia each have a different pet: a goldfish, a rabbit, a cat, a dog and a turtle. Lucas has a pet that barks. Hana has a furry animal. Fabienne has a furry animal. Leo owns a four-legged creature. Hana doesn't like cats.
Who has a dog?

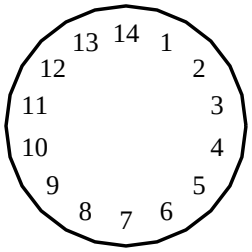
A. Leo ***B.*** Lucas ***C.*** Hana
D. Sofia ***E.*** Impossible to determine

3. Omar has the cards 1, 2 and 8. Omar must use every card exactly once, with +, –, ×, ÷ and brackets.
Which of these numbers can Omar get?

A. 22 ***B.*** 20 ***C.*** 19 ***D.*** 15 ***E.*** 13

Part B: Each question is worth 4 points

4. A round cake with the numbers 1 to 14 around its edge is cracked into 3 pieces. The sums of the numbers in the pieces are three consecutive numbers.



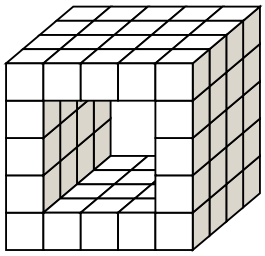
What is the largest of these sums?

A. 26 ***B.*** 33 ***C.*** 34 ***D.*** 35 ***E.*** 36

5. Alex has the cards 3, 7 and 8. Alex must use every card exactly once, with +, –, ×, ÷ and brackets. Which number can Alex *not* get?

A. 5 ***B.*** 12 ***C.*** 13 ***D.*** 15 ***E.*** 17

6. Sara built this tunnel out of identical cubes. The hole goes straight through from the front to the back. Sara wants to fill the hole completely, so that the tunnel becomes one solid block.



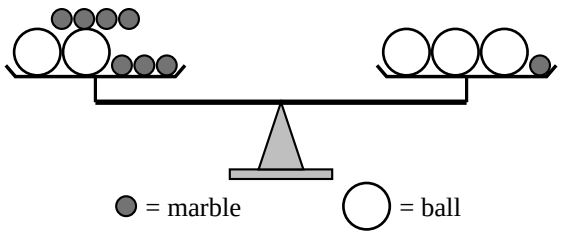
How many more cubes are needed?

A. 36 ***B.*** 35 ***C.*** 27 ***D.*** 12 ***E.*** 9

7. Zoe, Chloe, Ivan, Manon and Sofia each play a different instrument: the drum, the piano, the violin, the harp and the trumpet. Chloe presses keys. Sofia blows into the instrument. Ivan plays an instrument with strings. Manon uses a bow.
Which statement is true?

A. Zoe plays the violin ***B.*** Zoe plays the trumpet ***C.*** Zoe plays the piano
D. Sofia plays the violin ***E.*** Sofia plays the trumpet

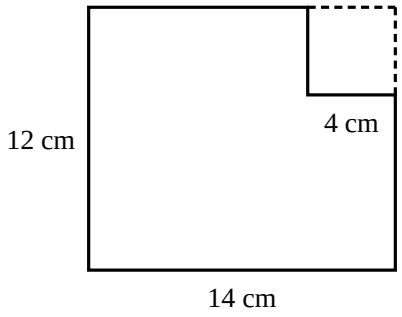
8. The balance is level: 2 balls and 7 marbles on one plate, 3 balls and 1 marble on the other.



One ball weighs the same as how many marbles?

A. 6 marbles ***B.*** 7 marbles ***C.*** 8 marbles ***D.*** 9 marbles ***E.*** 12 marbles

9. A rectangle is 14 cm long and 12 cm wide. A square with side 4 cm is cut off one corner.

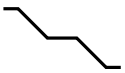
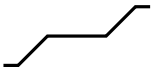


What is the perimeter of the remaining shape?

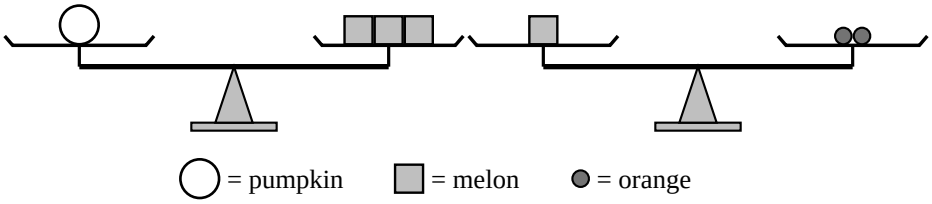
A. 68 cm ***B.*** 60 cm ***C.*** 52 cm ***D.*** 44 cm ***E.*** 36 cm

10. This is the outline of a town seen from the river at sunset.

Which piece **can** be part of the outline?

A.  ***B.*** ***C.*** ***D.***  ***E.***

11. The two balances are level. One pumpkin weighs as much as 3 melons, and one melon weighs as much as 2 oranges.

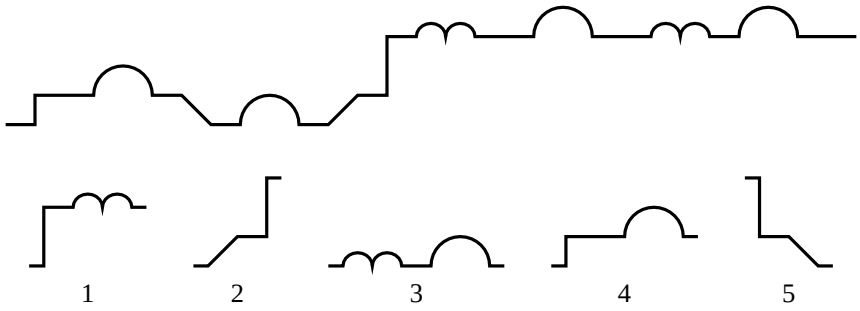


How many oranges weigh as much as 2 pumpkins?

A. 4 ***B.*** 10 ***C.*** 12 ***D.*** 14 ***E.*** 15

Part C: Each question is worth 5 points

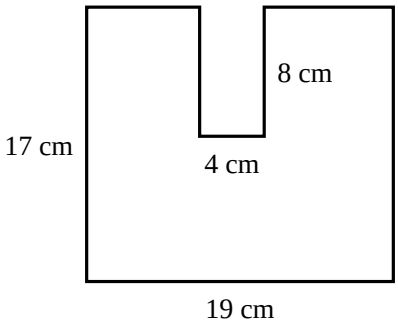
12. This is the outline of a town seen from the river at sunset.



How many of the pieces 1–5 below belong to the line?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

13. A 19 cm by 17 cm rectangle of cardboard has a rectangular notch 4 cm wide and 8 cm deep cut out of the middle of its top side.



What is the perimeter of the new shape?

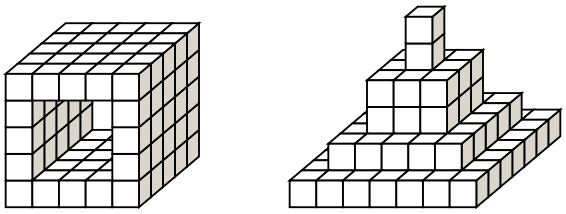
A. 88 cm ***B.*** 84 cm ***C.*** 80 cm ***D.*** 72 cm ***E.*** 68 cm

14. Ben, Ravi, Mia, Ivan and Emma were born in different months: June, April, January, March and July. Ben was not born in April. Emma was not born in July. Emma has a birthday in summer. Emma and Ivan both were born in a month that starts with J. Mia has a birthday in summer. Which statement is **not** true?

A. Mia was born in July ***B.*** Ravi was born in June ***C.*** Ravi was born in April

D. Emma was born in June ***E.*** Ivan was born in January

15. Leo has the tunnel on the left. Leo takes it apart to build the pyramid on the right, but there are not enough cubes. Nothing is hollow inside the pyramid.

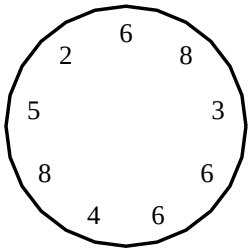


tunnel pyramid

How many more cubes does Leo need?

A. 4 ***B.*** 14 ***C.*** 16 ***D.*** 30 ***E.*** 78

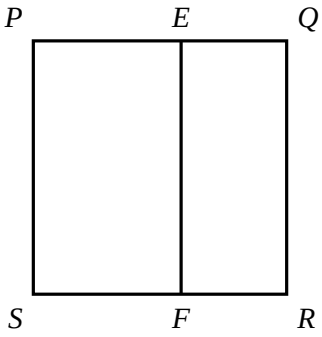
16. Numbers are written around a round plate as shown. Tom wants to break the plate with one straight line so that the two pieces have the same sum of numbers.



In how many different ways can he do it?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 5

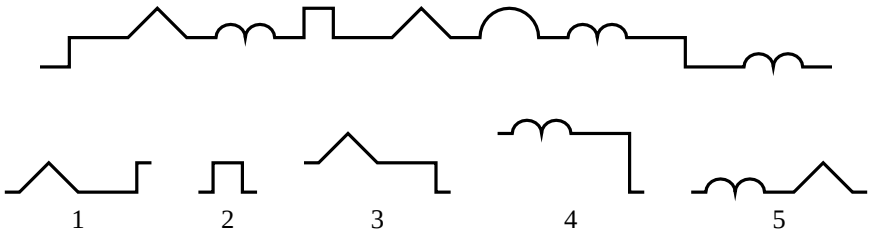
17. The square *PQRS* is cut into two rectangles. The perimeter of *PEFS* is 38 cm and the perimeter of *EQRF* is 34 cm.



What is the perimeter of the square *PQRS*?

A. 72 cm ***B.*** 48 cm ***C.*** 18 cm ***D.*** 12 cm ***E.*** 4 cm

18. Lena cut a strip of paper along this line.



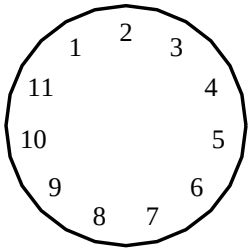
How many of the pieces 1–5 below belong to the line?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 5

19. Fred has the cards 1, 3, 5 and 9. Using every card exactly once, with +, –, ×, ÷ and brackets, Fred wants to get the biggest possible number.
What is the biggest number Fred can get?

A. 218 ***B.*** 217 ***C.*** 184 ***D.*** 180 ***E.*** 18

20. A round plate with the numbers 1 to 11 around its edge is broken by one straight crack into two pieces. The sums of the numbers on the two pieces are equal.



Where does the crack go?

A. between 3 and 4, and between 9 and 10
B. between 6 and 7, and between 11 and 1
C. between 4 and 5, and between 10 and 11
D. between 1 and 2, and between 9 and 10
E. between 2 and 3, and between 8 and 9

Answer Key

1.	**A**	**2.**	**B**	**3.**	**D**	**4.**	**E**	**5.**	**D**	**6.**	**A**
7.	**E**	**8.**	**A**	**9.**	**C**	**10.**	**B**	**11.**	**C**	**12.**	**E**
13.	**A**	**14.**	**B**	**15.**	**B**	**16.**	**B**	**17.**	**B**	**18.**	**B**
19.	**D**	**20.**	**E**								

Solutions

1. A (5 plums) — Take 1 pineapple off both plates: 5 plums balance 2 – 1 = 1 pineapple. So one pineapple = 5 plums. [2002 Q9 · one_more_balances]

2. B (Lucas) — The only possible matching: Lucas – dog; Hana – rabbit; Leo – turtle; Fabienne – cat; Sofia – goldfish. [2002 Q18 · who_has]

3. D (15) — 2 × 8 – 1 = 15. None of the other numbers can be made. [2002 Q21 · which_can]

4. E (36) — All the numbers add up to 105. Three consecutive numbers that add up to 105: 34 + 35 + 36 = 105. The largest is 36. [2002 Q11 · largest_sum]

5. D (15) — (7 + 8) ÷ 3 = 5; 7 + 8 – 3 = 12; 3 × 7 – 8 = 13; 3 × 8 – 7 = 17. But with 3, 7 and 8 nothing gives 15 (the closest are 13 = 3 × 7 – 8 and 17 = 3 × 8 – 7). [2002 Q21 · which_cannot]

6. A (36) — The hole is 3 × 3 = 9 cubes in the front and goes 4 cubes back: 9 × 4 = 36. [2002 Q20 · fill_the_hole]

7. E (Sofia plays the trumpet) — The only possible matching: Zoe – drum; Chloe – piano; Ivan – harp; Manon – violin; Sofia – trumpet. [2002 Q18 · which_true]

8. A (6 marbles) — Take 2 balls and 1 marble off both plates. Left: 1 ball = 6 marbles. [2002 Q9 · both_sides]

9. C (52 cm) — The two new edges (4 cm and 4 cm) replace the two removed pieces of the outer edge, which were also 4 cm and 4 cm. So the perimeter is the same as the rectangle's: 2 × (14 + 12) = 52 cm. [2002 Q6 · corner_cut]

10. B — Only this piece is found in the line. The others don't fit — A: it is a mirror image of a part of the line; C: it steps up where the line steps down; D: its flat part between the two features has the wrong length; E: it has the right shapes but in the wrong order. [2002 Q7 · can_belong]

11. C (12) — One pumpkin = 3 melons = 3 × 2 = 6 oranges; 2 pumpkins = 2 × 6 = 12 oranges. [2002 Q9 · chain]

12. E (4) — Pieces 1, 2, 3, 4 fit. Not: 5 (it is a mirror image of a part of the line). [2002 Q7 · how_many_fit]

13. A (88 cm) — The bottom of the notch (4 cm) replaces the 4 cm taken out of the top edge, and the two sides of the notch add 2 × 8 cm. Perimeter = 2 × (19 + 17) + 2 × 8 = 88 cm. [2002 Q6 · notch]

14. B (Ravi was born in June) — The only possible matching: Ben – March; Ravi – April; Mia – July; Ivan – January; Emma – June. [2002 Q18 · which_not_true]

15. B (14) — Tunnel: 16 × 5 = 80 cubes. Pyramid: 7 × 7 + 5 × 5 + 3 × 3 × 2 + 1 × 1 × 2 = 94. Missing: 94 – 80 = 14. [2002 Q20 · not_enough]

16. B (1) — Each half must have 24. The ways: 6+6+4+8 | 5+2+6+8+3. [2002 Q11 · count_ways]

17. B (48 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 38 + 34 = 72 cm = 6 sides. Side = 72 ÷ 6 = 12 cm, perimeter 4 × 12 = 48 cm. [2002 Q6 · two_perimeters]

18. B (1) — Only piece 2 fits. Not: 1 (it has the right shapes but in the wrong order); 3 (it is a mirror image of a part of the line); 4 (its step is taller than in the line); 5 (it is a mirror image of a part of the line). [2002 Q7 · how_many_fit]

19. D (180) — (1 + 3) × 5 × 9 = 180. Multiplying by 1 changes nothing, so it is better to add the 1 to the smallest other card first. [2002 Q21 · biggest]

20. E (between 2 and 3, and between 8 and 9) — Total 66, so each piece has 33: 3+4+5+6+7+8 = 33 and 9+10+11+1+2 = 33. [2002 Q11 · straight_crack]